Roll No. ....

TME–302

B. TECH. (ME) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

MATERIAL SCIENCE

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1\. (a) What is crystallography ? What are the different crystal structures ? Explain in detail. (CO1/CO2)

OR

(b) Volume of a FCC unit cell is 67.42×10^{-30} m^3. Calculate the atomic diameter of its atom. (CO1/CO2)

2\. (a) Derive the expression for relation atomic radius and lattice constant for a simple cubic, body centered cubic, and face centered cubic structure. (CO1/CO2)

OR

(b) Obtain the miller indices of a plane whose intercept are a, $b/2$, $3c$ on x, y and z axes respectively in a simple cubic unit cell. (CO1/CO2)

3\. (a) Detail out the procedure to obtain Miller indices in Cubic Structure. (CO1/CO2)

OR

(b) Calculate the spacing between (1 1 1), (1 1 0) and (1 0 0) place, in a crystal of NaCl whose lattice parameter is 5.64 A. (CO1/CO2)

4. (a) What are the Defects ? How are they classified ? Explain any *two* defects in detail. (CO2/CO3)

OR

(b) How are mechanical properties important to a design engineer ? Explain any *five* mechanical properties. (CO2/CO3)

5. (a) Draw Stress-Strain diagram for mild steel and explain all the important points on it. (CO3/CO4)

OR

(b) Calculate the of alloy in Pb-Sn system at which the fraction of total α is 2.5 times the fraction of β phase at 182°C. (CO3/CO4)

Roll No. ....

TMA–303

B. TECH. (ME) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

ENGINEERING MATHEMATICS–III

Time : 1:30 Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub questions.

(ii) Each sub-question carries 10 marks.

1. (a) Show that the function :

$$f(z) = u + iv$$

where

$$f(z) = \begin{cases} \dfrac{x^2 y^5 (x+iy)}{x^4 + y^{10}}, & z \neq 0 \\ 0, & z = 0 \end{cases}$$

is not analytic at $z = 0$ although Cauchy-Riemann equations are satisfied at $z = 0$. (CO1)

OR

(b) Write down the statement of Cauchy's integral formula, and hence to evaluate the integral $\oint \frac{z}{z^2 - 3z + 2}$. (CO2)

P. T. O.

2. (a) Evaluate $\int_{1-i}^{2+i}(2x+iy+1)dz$ along the straight line $1-i$ and $2+i$.

(CO2)

OR

(b) Define the Singularities of Complex functions with the help of an example. (CO2)

3. (a) Define the harmonic function. Check the real part of : (CO1)

$$f(z)=u(x,y)+iv(x, y)$$

where

$$u(x,y)=x^2-y^2-2xy-2x+3y$$

is harmonic. If yes, then find the analytic function $f(z)$.

OR

(b) Find the value of the integral $\int_C (x+y)\,dx+x^2y\,dy$:

(i) along $y=x^2$ having (0, 0) and (3, 9) end points

(ii) along $y=3x$ between the same points. (CO2)

4. (a) Write down the statement of Laurent's theorem. Find the Laurent's series expansion of the function $f(z)=\dfrac{z^2-1}{z^2+5z+6}$ in the domain $2<|z|<3$. (CO2)

OR

(b) Using Cauchy's integral formula to compute the value of the integral : (CO2)

$$\oint_C \frac{\sin \pi z^2+\cos \pi z^2}{(z-1)(z-2)}dz$$

where

$$C:|x+iy|=3$$

5. (a) Define the analytic function, and hence to construct the analytic function $f(z)$ using Milne-Thomson's method where the real part of function $f(z)$ is : (CO1)

$$u(x, y) = e^{-x}[(x^2 - y^2)\cos y + 2xy \sin y]$$

OR

(b) Define the Translation, Magnification, Rotation, and Inversion with the help of example. (CO3)

5. (a) Define the analytic function, and hence to construct the analytic function $f(z)$ using Milne-Thomson's method where the real part of function $f(z)$ is : (CO1)

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OR

(b) Define the Translation, Magnification, Rotation, and Inversion with the help of example. (CO3)

Roll No. ....

TME–304

B. TECH. (ME) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

BASIC THERMODYNAMICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) (i) Define thermodynamic systems. Classify and define them with examples. (CO1)

(ii) State and differentiate between extensive, intensive and specific properties.

OR

(b) (i) State and explain zeroth law of thermodynamics used for temperature measurement. (CO1)

(ii) Define : thermodynamic process, cycle, state/point function and path function.

P. T. O.

2. (a) In a reversible non-flow process, the work is done by a substance in accordance with $V=\frac{2.8}{P}m^3$, where p is the pressure in bar. Find the work done on or by system as pressure increases from 0.7 bar to 7 bar. (CO1)

OR

(b) Derive an expression for displacement work in an isothermal process and in an adiabatic process. (CO1)

3. (a) State the first law of thermodynamics for a closed system undergoing a cycle. Write the consequences of this law and prove that heat transfer is a path function. (CO2)

OR

(b) A gas undergoes a thermodynamic cycle consisting of three processes :

Process 1-2 : compression with pV = constant, from $p_1 = 1$ bar, $V_1 = 1.6$ m^3 to $V_2 = 0.2\ m^{31}$, $U_2 - U_1 = 0$

Process 2-3 : constant pressure to V3 = V1

Process 3-1 : constant volume, UI- U3 = – 3549 kJ

There are no significant changes in kinetic or potential energy. Determine the heat transfer and work for Process 2-3, and net work transfer in kJ. (CO2)

4. (a) 3 kg of air at a pressure of 150 kPa and temperature 360 K is compressed polytropically to 750 kPa according to law $pV^{1.2} = C$. The gas is then cooled to initial temperature at constant pressure. The air is then expanded at constant temperature till it reaches original pressure of 150 kPa. Draw the cycle on p-V diagram and determine the net work transfer and heat transfer. (CO1/CO2)

OR

(b) What is a steady flow process ? Derive an equation of SFEE. Also, explain how this expression can be used for a turbine and a nozzle.

(CO2)

4. (a) Air at 600 kPa and 500 K enters an adiabatic nozzle that has an inlet-to-exit area ratio of 2 :1 with a velocity of 120 m/s and leaves with a velocity of 380 m/s. Determine (i) the exit temperature and (ii) the exit pressure of the air. (CO2)

OR

(b) Steam enters a turbine with a velocity of 40 m/s and specific enthalpy of 2500 kJ/kg; and leaves with a velocity of 90 m/s and specific enthalpy of 2030 kJ/kg. Heat losses from the turbine to surroundings are 240 kJ/min and the steam-flow rate is 5040 kg/h. Neglect the change of potential energy. Find the power developed by the turbine. (CO2)

Roll No. ....

TME–305

B. TECH. (ME) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

MANUFACTURING PROCESSES–I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Find the velocity of molten metal the gate, volume rate of flow and time require to fill the mould, if sprue length if 40 cm and cross section area at the base of sprue is 8 cm^2. The volume of mould is 2000 cm^3.

(CO1/CO1)

OR

(b) Find time of cooling a casting take of spherical in shape.

Given : radius and height (thickness) of cylinder are 280 mm and 40 mm respectively. Mold constant as per Chvorinov's Rule $C_m = 3.5$ sec/mm^2.

(CO1/CO1)

2. (a) Find the net buoyancy force on core if height and diameter of core is given as 40 and 60 mm respectively. The density of core and molten metal is given as 1100 kg/m^3 and 2200 kg/m^3. Assume gravitational constant as standard value.

(CO1/CO1)

P. T. O.

OR

(b) Calculate the permeability number for a standard specimen of height 5.08 cm. It takes 120 seconds to allow 2100 cm^3 of air to pass through under a pressure difference of 9 gm/cm^3. (CO1/CO1)

3. (a) What is use of riser in sand casting ? Explain in brief. (CO1/CO1)

OR

(b) Explain any *three* types of patterns used in casting. What is the difference between permanent and temporary mould ? (CO1/CO1)

4. (a) Derive the equation for time taken by the metal to solidify in bottom casting. (CO1/CO1)

OR

(b) Draw neat and clean diagram and explain the centrifugal casting. (CO1/CO1)

5. (a) What is forging ? Write and explain any *two* applications of forging. (CO2/CO2)

OR

(b) Briefly explain drop forging with a neat diagram. (CO2/CO2)

Roll No. ....

TME–306

B. TECH. (ME) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

ENGINEERING MECHANICS

Time : 1½ Hours

Maximum Marks : 50

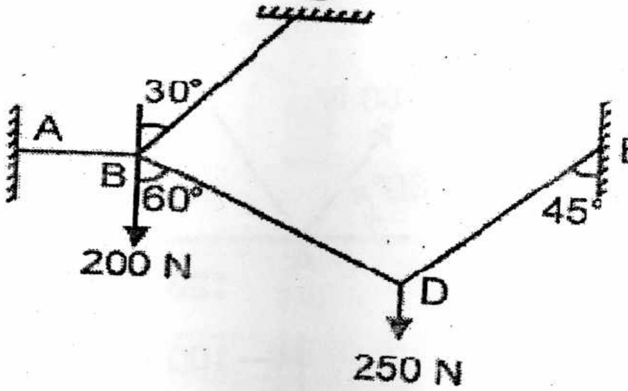
Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) State and explain Varignon's theorem of moments. (CO1, CO2)

OR

(b) A system of connected flexible cables shown in Figure is supporting two vertical forces 200 N and 250 N at points B and D. Determine the forces in various segments of the cable. (CO1, CO2)

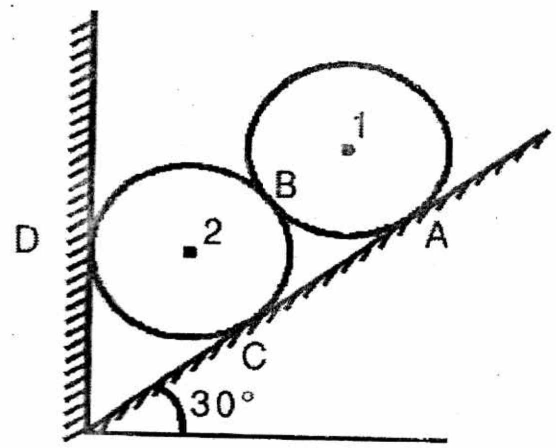


P. T. O.

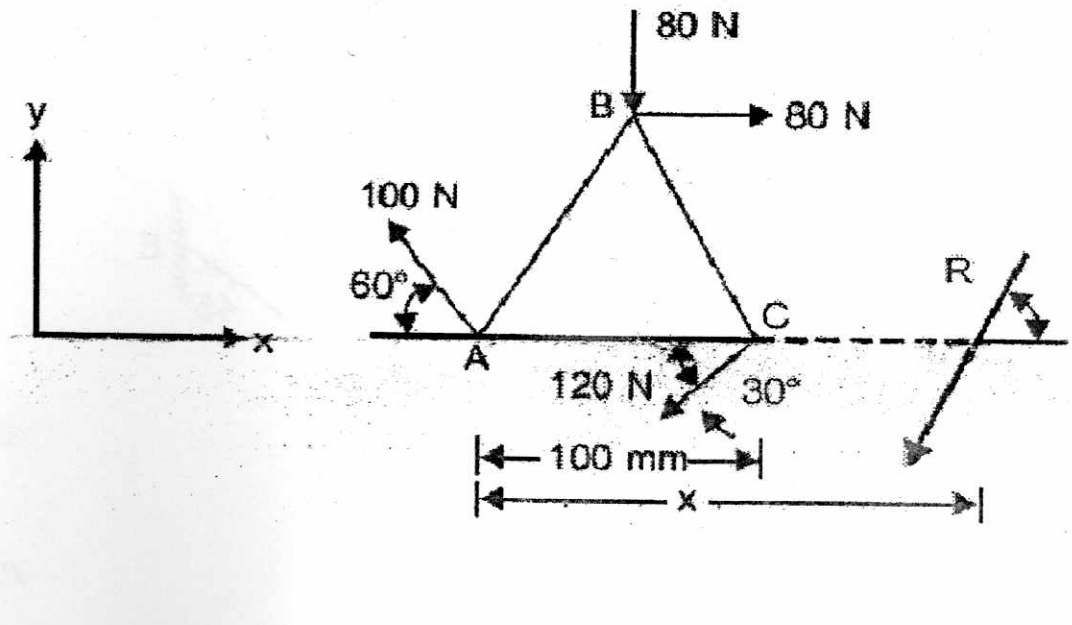
2. (a) State Parallelogram law of forces and drive for resultant. (CO1, CO2)

OR

(b) Two identical rollers, each of weights 500 N are supported by an inclined plane and a vertical wall as shown in the figure, Assuming smooth surfaces, find the reactions induced at the points of supports A, B, C, and D. (CO1, CO2)

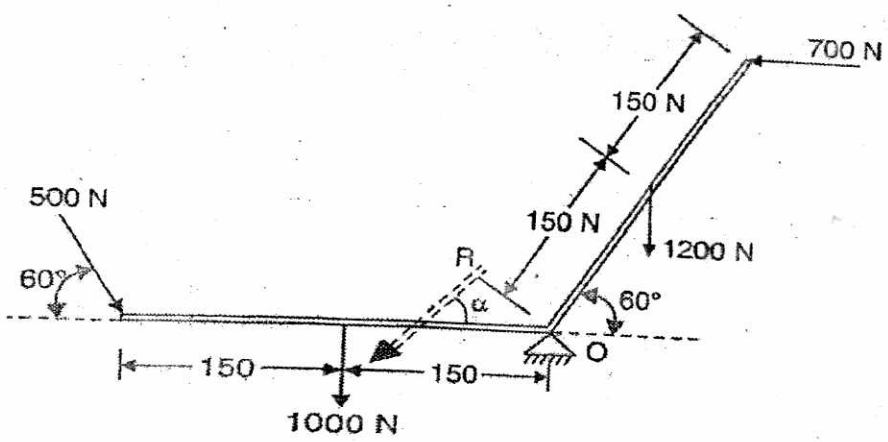


3. (a) Find magnitude, direction and point of application of the resultant of the force system shown in Figure, acting on a lamina of equilateral triangular shape. (CO1, CO2)



OR

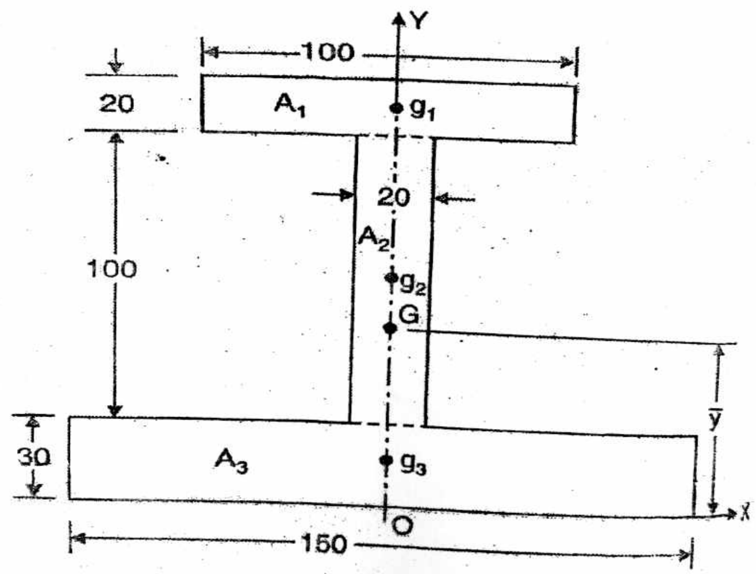
(b) The system of forces acting on a bell crank as shown in figure. Determine the magnitude, direction and point of application on the resultant. (CO1, CO2)



4. (a) Locate the centroid of a semicircle from its diametral axis. (CO3)

OR

(b) Locate the centroid of the I-section shown in the figure. (CO3)

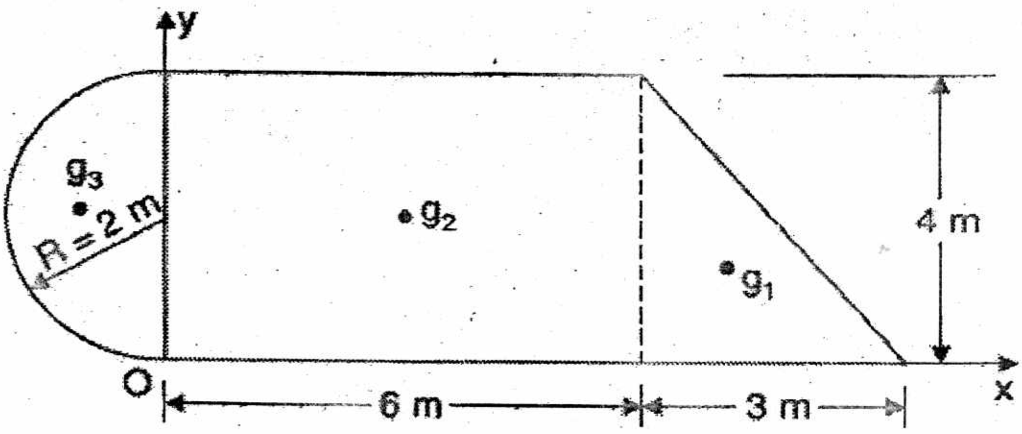


All dimensions in mm

5. (a) Locate the centroid of sector of a circle from its centre. (CO3)

OR

(b) Determine the centroid of the area shown in Figure, with respect to the axis shown. (CO3)



Roll No. ....

TME–308

B. TECH. (ME) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

MANAGEMENT OF INTELLECTUAL PROPERTY RIGHTS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define the concept of a patent and elaborate on the term of a patent in the Indian patent system. (CO1/CO3)

OR

(b) Identify the governing legislation for patents In India and examine whether an Indian patent provides worldwide protection. (CO1/CO3)

2. (a) Explain what is eligible for patent protection and outline the criteria that Inventions must meet to be patentable. (CO1/CO4)

OR

(b) Enumerate the types of inventions that are not patentable in India referencing the relevant sections of the Patent Act, 1970. (CO1/CO4)

3. (a) Provide an overview of the various categories of patent applications and offer brief explanations for each. (CO2/CO3)

P. T. O.

OR

(b) Detail the specific forms that need to be completed when filing a patent application in India. (CO2/CO3)

4. (a) Investigate whether the Patents Act includes any provisions for early publication and provide an explanation if such provisions exist. (CO4)

OR

(b) Analyze the significant principles and provisions of the Berne Convention, which safeguards literary and artistic works. (CO4)

5. (a) Explore the qualifications and disqualifications for individuals seeking to register as patent agents and discuss their rights. (CO4/CO1)

OR

(b) Describe scenarios in which the valuation of intellectual property is crucial for an organization's decision-making process. (CO4/CO1)

Roll No. ....

TRA–301

B. TECH. (MECHANICAL ENGG./ROBOTICS AND AUTOMATION) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

MICROPROCESSOR AND MICROCONTROLLER

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Write counting from 0 to 32 in :

   (i) Octal

   (ii) Hexadecimal (CO1)

   OR

   (b) Determine the range of :

   (i) signed magnitude representation

   (ii) l's complement signed representation for 4 bit numbers (CO1)

2. (a) Convert the following numbers in decimal :

   (i) 24H

   (ii) $(37)_8$

   (iii) $(11010101)_2$

   (iv) $(6A)_{16}$ (CO1)

P. T. O.

12|10|23
2nd shift

OR

(b) Find l's and 2's complement of the following binary numbers :

(i) 1011010

(ii) 11010100

(iii) 111001 (CO1)

3. (a) Perform all the four arithmetic operations using l's complement form for decimal numbers 19 and 14. (CO1/CO2)

OR

(b) What is the need of coding ? Write a short note on BCD code. (CO1/CO2)

4. (a) Determine the range of 8085 microprocessor data and addressing capacity. (CO1/CO2)

OR

(b) Write a short note on memory devices. (CO1/CO2)

5. (a) Explain 8085 microprocessor programming model with the help of a diagram. (CO2)

OR

(b) Explain microprocessor based system with bus architecture with the help of a block diagram and discuss its various components. (CO2)

Roll No. ....

TAE–301

B. TECH. (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

AUTOMOTIVE SYSTEMS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define automobiles and elaborate on their significance, necessity, and diverse applications. (CO1)

OR

(b) Categorize different types of automobiles while linking each type to its specific field of use. (CO1)

2. (a) Provide a list of at least *three* special purpose vehicles, along with explanations of their intended uses. (CO1/CO4)

OR

(b) Examine the reasons behind government initiatives to discontinue the production of internal combustion engines. (CO1/CO4)

3. (a) Compare and contrast 2-stroke and 4-stroke engines, using a spark ignition engine as an example. (CO4/CO3)

P. T. O.

OR

(b) Clarify the function of an automotive chassis and offer a concise classification. (CO4/CO3)

4. (a) Explore the essential role of transmissions In vehicles with internal combustion engines, substantiated with appropriate graphs or diagrams. (CO3)

OR

(b) Enumerate the advantages and disadvantages associated with manual transmissions. (CO3)

5. (a) Offer an explanation of how a carburetor functions, supported by a suitable diagram. (CO5/CO2)

OR

(b) Compile a list of various electrical and electronic components found in automobiles, along with their intended purposes. (CO5/CO2)

Roll No. ....

TMS–301

B. TECH. (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

PRODUCT DESIGN AND DEVELOPMENT

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain in detail the concept of 'Product Definition'. Also discuss what do you understand by 'Multifunctional Products'. (CO1)

OR

(b) Discuss in detail about the 'Stage-gate process' of Product development. (CO1)

2. (a) Write in detail about the concept of 'Requirement Gathering and Management' with respect to product development. (CO1/CO2)

OR

(b) Explain in detail about the concept of 'Product architecture'. Also state the differences between Modular and Integral architecture. (CO1/CO2)

3. (a) Discuss in detail about the concept of 'Teardown and Benchmarking'. (CO1)

P. T. O.

OR

(b) Write in detail about the concept of 'Advanced Manufacturing'. (CO1)

4. (a) Discuss in detail about the concept of 'Idea Generation'. (CO2/CO3)

OR

(b) Briefly explain the concept of 'Product Target'. (CO2/CO3)

5. (a) Explain in detail about the concept of 'Creativity' and its applications in Problem solving. (CO3)

OR

(b) Write in detail about the significance of 'Conceptual Decomposition'. (CO3)